Modelling animal conflicts using (evolutionary) game theory

Shikhara

M14A16: Theoretische Evolutionsbiologie und Ökologie

King Cobras: Extremely venomous, specialise on eating other snakes

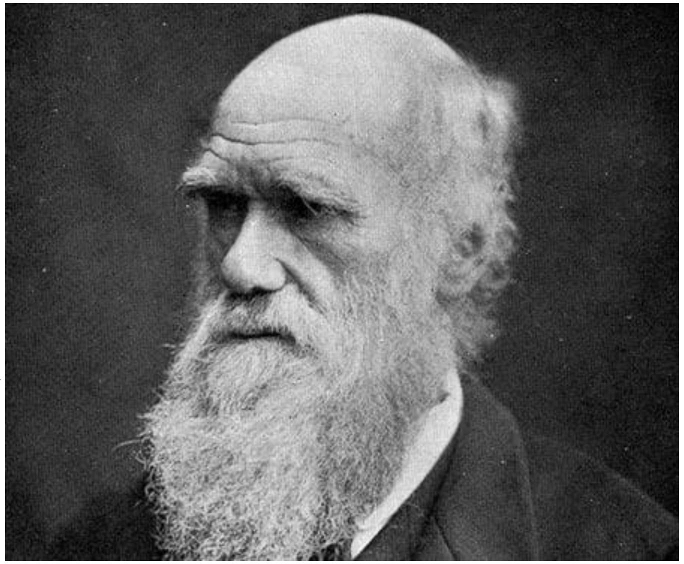
From Wikipedia:

A king cobra's bite, and subsequent envenomation, is an immediate medical emergency in humans or domesticated animals, as, if not treated as soon as possible, death can occur in as little as 30 minutes.[40][58]

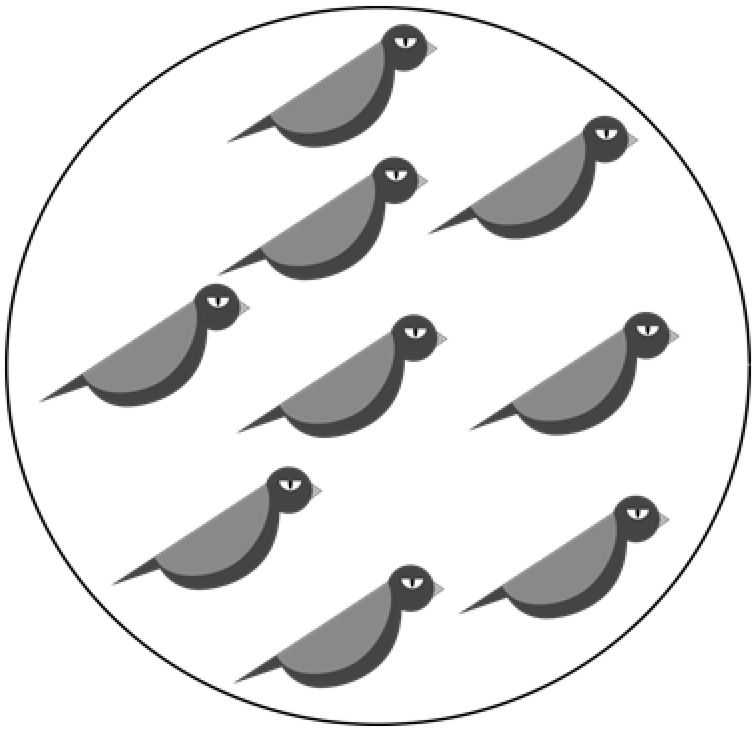
It can deliver up to 420 mg venom in dry weight (400–600 mg overall) per bite,[61] with a LD_{50} toxicity in mice of 1.28 mg/kg through intravenous injection,[62] 1.5 to 1.7 mg/kg through subcutaneous injection,[63] and 1.644 mg/kg through intraperitoneal

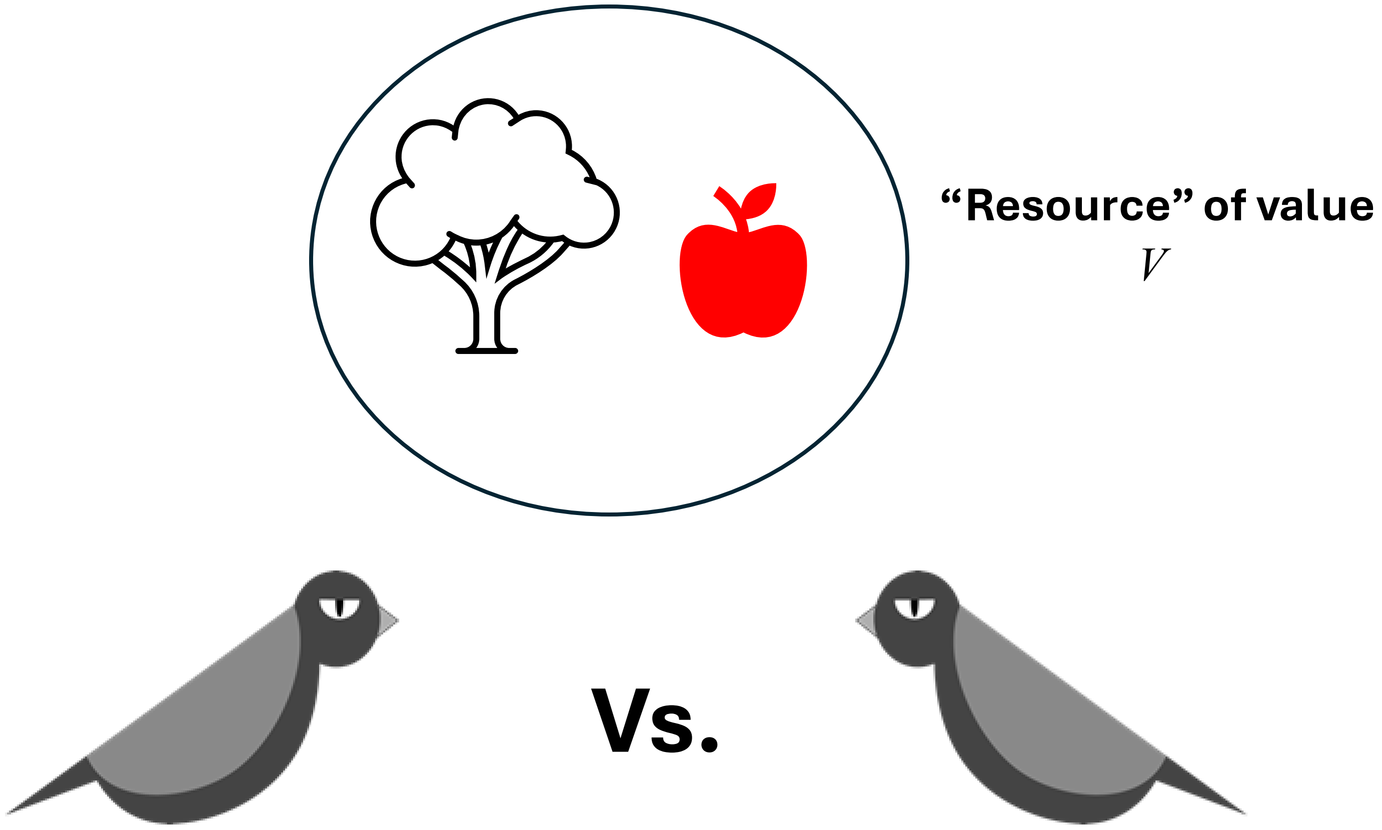
Ritualised combat displays to determine the winner of territorial disputes

wtf

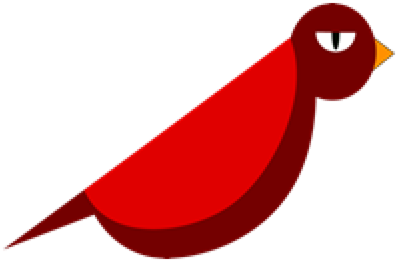


Our goal: try to understand how such ‘restraint’ during combat may evolve through natural selection



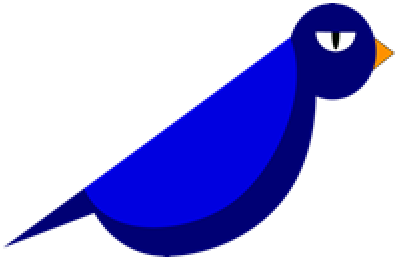


A simple model: two kinds of ‘strategies’



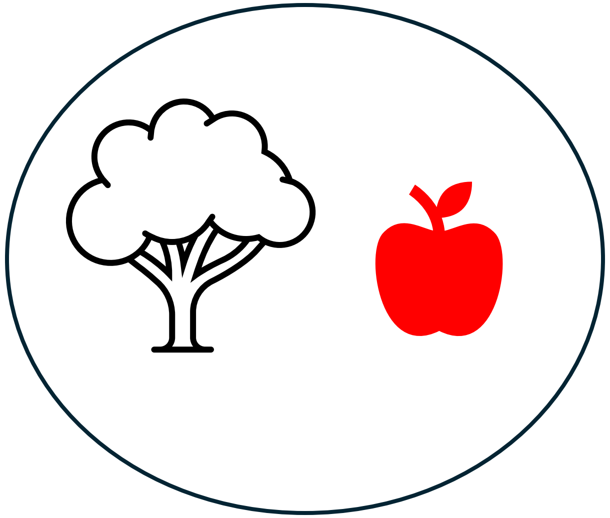
“Hawk”: Aggressive, uses all its weaponry (beak, claws, fangs, venom, ...), always fights and injures its opponent.

Being injured by the weaponry incurs a cost C

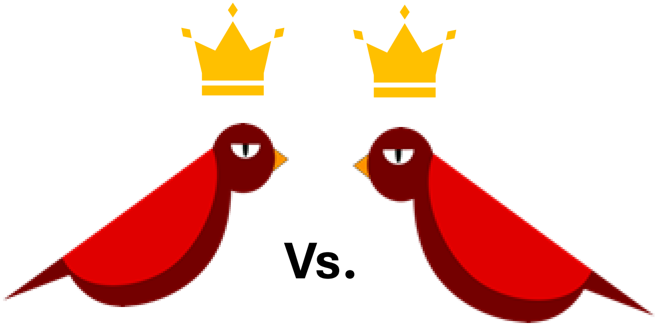


“Dove”: Engages in ritualised displays. If the opponent fights with weaponry, retreats and gives up the resource before any potential for injury

Which interactions are possible?

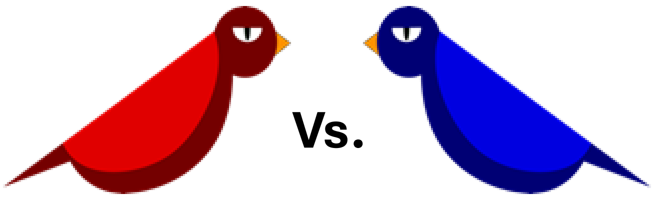


"Resource" of value
V

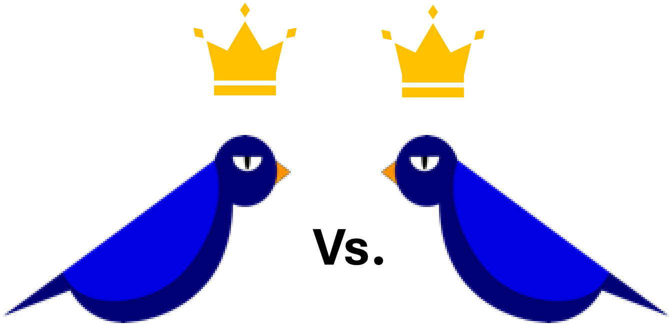


V $-C$

$-C$ V



V 0

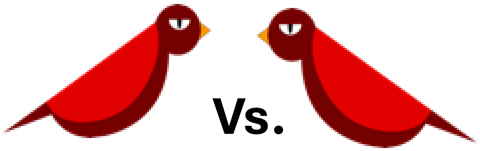


V 0

0 V

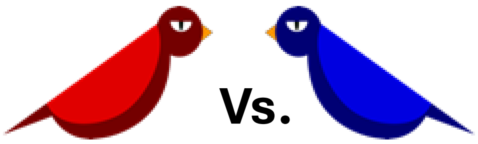
A more concise representation of the interactions

Opponent → / Focal individual ↓	Hawk 	Dove 
Hawk 	$\frac{1}{2}V - \frac{1}{2}C$	
Dove 		

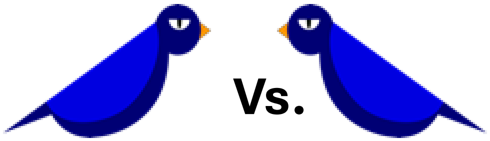


V $\quad$ $-C$

$-C$ $\quad$ V



V $\quad$ 0

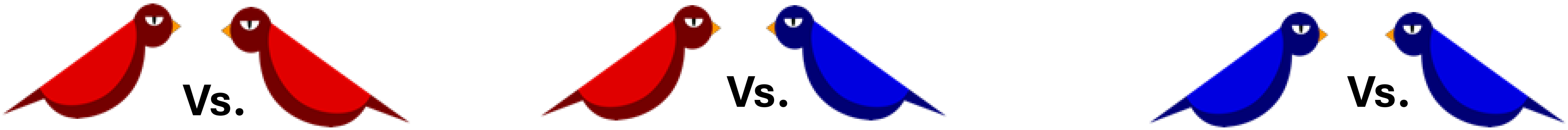


V $\quad$ 0

0 $\quad$ V

A more concise representation of the interactions

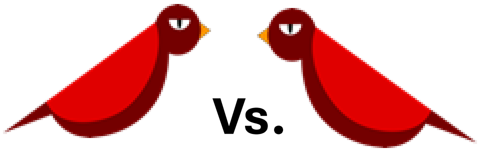
Focal individual ↓ \ Opponent →	Hawk	Dove
Hawk	$\dfrac{V-C}{2}$	V
Dove	0	$\dfrac{1}{2}V + \dfrac{1}{2}\cdot 0$



V	$-C$			V	0
$-C$	V	V	0	0	V

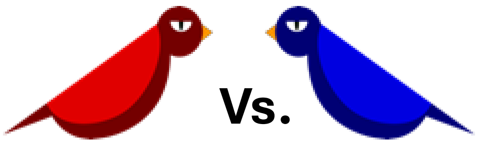
A more concise representation of the interactions

Opponent → / Focal individual ↓	Hawk 	Dove 
Hawk 	$\frac{V-C}{2}$	V
Dove 	0	$\frac{V}{2}$

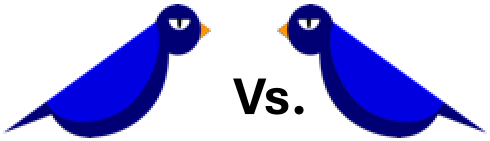


V $-C$

$-C$ V



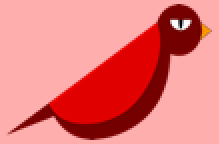
V 0

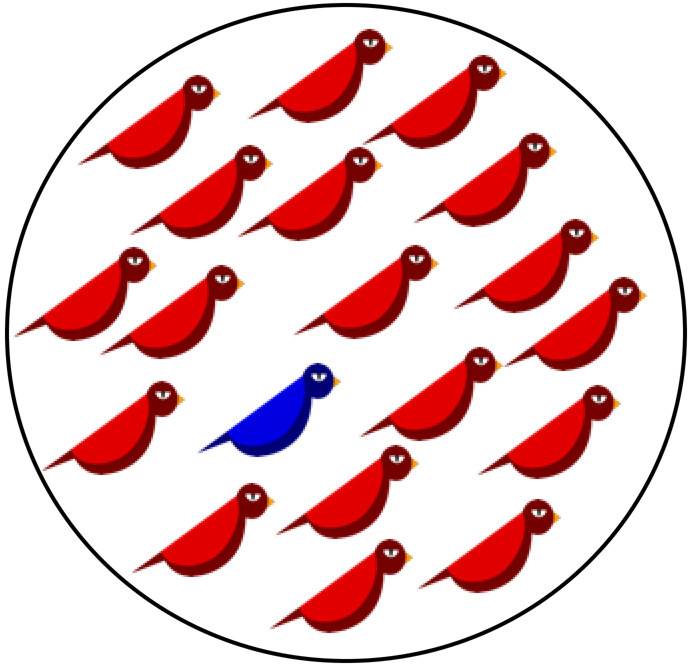


V 0

0 V

Will a rare Dove spread in a Hawk population?

Opponent → / Focal individual ↓	Hawk 	Dove 
Hawk 	$\frac{V-C}{2}$	V
Dove 	0	$\frac{V}{2}$



$$\text{Average hawk fitness} \approx \frac{V-C}{2}$$

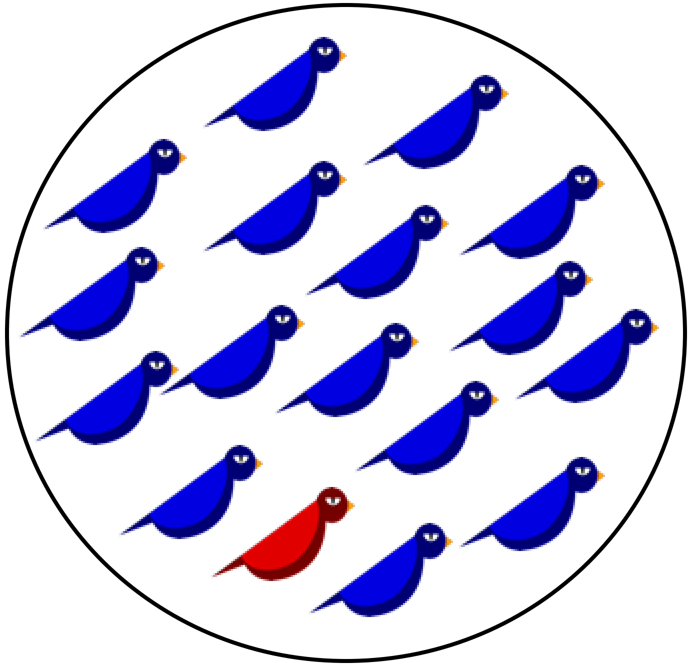
$$\text{Average dove fitness} \approx 0$$

The mutant Dove is eliminated by natural selection if $\underbrace{\frac{V-C}{2}}_{\substack{H-H \\ \text{outcome}}} > \underbrace{0}_{\substack{D-H \\ \text{outcome}}}$

In this case, we say Hawk is an **Evolutionarily Stable Strategy (ESS).**

Can Dove be an ESS?

Focal individual ↓ / Opponent →	Hawk	Dove
Hawk	$\dfrac{V-C}{2}$	V
Dove	0	$\dfrac{V}{2}$



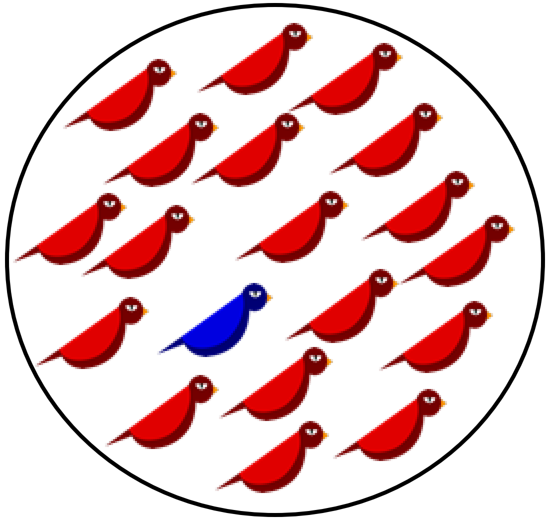
$$\text{Average hawk fitness} \approx V$$

$$\text{Average dove fitness} \approx \frac{V}{2}$$

The mutant Hawk is eliminated by natural selection if $\underbrace{\frac{V}{2}}_{\substack{D\text{–}D\\ \text{outcome}}} > \underbrace{V}_{\substack{H\text{–}D\\ \text{outcome}}}$ (never true!)

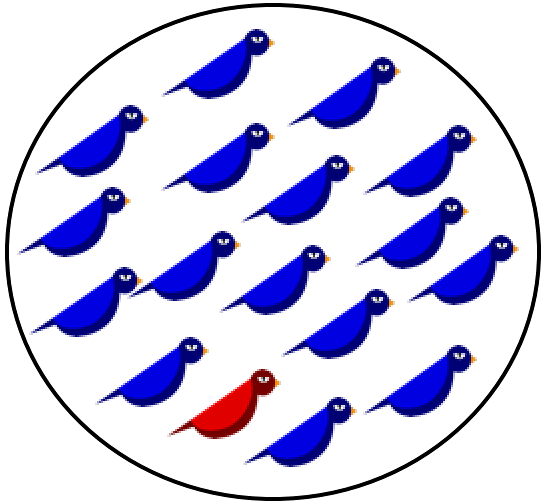
When $V > C$, we don't expect to see Doves

Opponent → / Focal individual ↓	Hawk	Dove
Hawk	$\frac{V-C}{2}$	V
Dove	0	$\frac{V}{2}$



A rare mutant Dove cannot spread via natural selection in a population of Hawks

(Hawk **is stable** against Dove/ Dove cannot '**invade**' hawk)



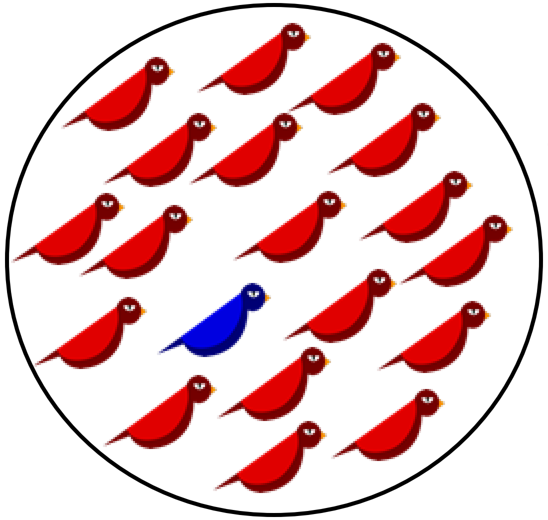
A rare mutant Hawk will spread via natural selection in a population of Doves.

(Dove **is unstable** against Hawk/ Hawk will '**invade**' Dove)

When $V < C$, neither strategy is an ESS

(Value of resource < Cost of injury)

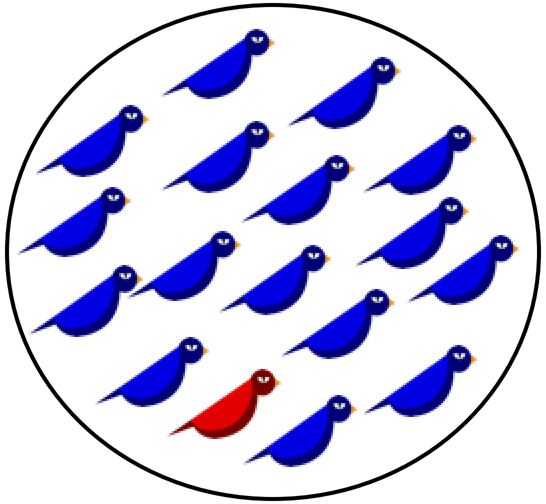
Focal individual ↓ \ Opponent →	Hawk 	Dove 
Hawk 	$\frac{V-C}{2}$	V
Dove 	0	$\frac{V}{2}$



Average hawk fitness $\approx \frac{V-C}{2}$

Average dove fitness ≈ 0

A rare mutant Dove will spread via natural selection in a population of Hawks



Average hawk fitness $\approx V$

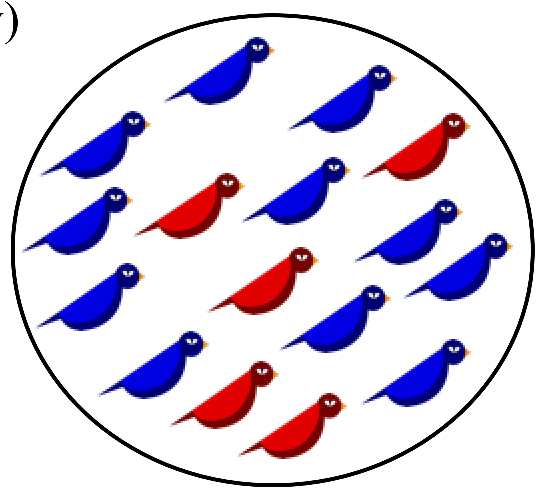
Average dove fitness $\approx \frac{V}{2}$

A rare mutant Hawk will spread via natural selection in a population of Doves

What happens when $V < C$?

(Value of resource < Cost of injury)

Opponent → / Focal individual ↓	Hawk 	Dove 
Hawk 	$\frac{V-C}{2}$	V
Dove 	0	$\frac{V}{2}$



Let $p :=$ proportion of Hawks in the population
$1 - p =$ proportion of Doves in the population

$$\text{Avg fitness of Hawks } = \Pr(H - H \text{ interaction}) \underbrace{\frac{V-C}{2}}_{\substack{H-H \\ \text{Outcome}}} + \Pr(H - D \text{ interaction}) \underbrace{V}_{\substack{H-D \\ \text{Outcome}}}$$

$$= p\left(\frac{V-C}{2}\right) + (1-p)V$$

$$\text{Avg fitness of Doves } = \Pr(D - H \text{ interaction}) \underbrace{0}_{\substack{D-H \\ \text{Outcome}}} + \Pr(D - D \text{ interaction}) \underbrace{\frac{V}{2}}_{\substack{D-D \\ \text{Outcome}}}$$

$$= (1-p)\frac{V}{2}$$

What happens when $V < C$?

(Value of resource < Cost of injury)

Opponent → / Focal individual ↓	Hawk 	Dove 
Hawk 	$\dfrac{V-C}{2}$	V
Dove 	0	$\dfrac{V}{2}$

Let $p :=$ proportion of Hawks in the population
$1 - p =$ proportion of Doves in the population

$$\text{Avg fitness of Hawks} \quad = p\left(\frac{V-C}{2}\right) + (1-p)V$$

$$\text{Avg fitness of Doves} \quad = (1-p)\frac{V}{2}$$

When is Avg Hawk fitness > Avg Dove fitness ?

$$p\left(\frac{V-C}{2}\right) + (1-p)V > (1-p)\frac{V}{2} \Rightarrow p < \frac{V}{C}$$

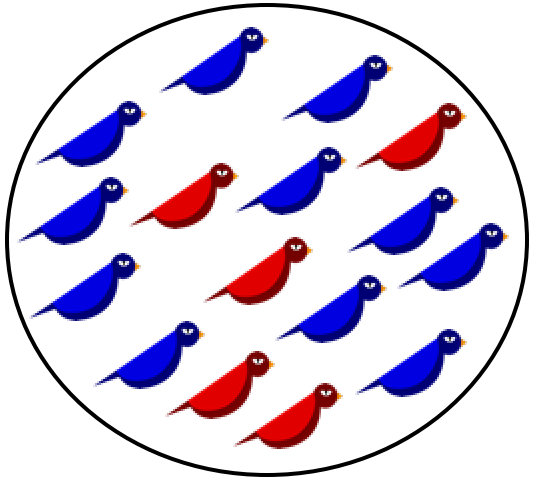
At $p = \dfrac{V}{C}$, Avg Hawk fitness = Avg Dove fitness

Thus a population with a proportion of $\dfrac{V}{C}$ Hawks and $\left(1 - \dfrac{V}{C}\right)$ Doves is at equilibrium.

What happens when $V < C$?

(Value of resource < Cost of injury)

Opponent → / Focal individual ↓	Hawk 	Dove 
Hawk 	$\frac{V-C}{2}$	V
Dove 	0	$\frac{V}{2}$



A population with a proportion V/C Hawks (aggressive, using all weaponry) and remaining individuals Doves (displaying/using ritualised combat) is at equilibrium for natural selection. We call such a population a ‘mixed ESS’.

The connection to dynamics

a population with a proportion of $\frac{V}{C}$ Hawks and $\left(1-\frac{V}{C}\right)$ Doves is at equilibrium.

Focal individual ↓ \ Opponent →	Hawk 	Dove 
Hawk 	$\frac{V-C}{2}$	V
Dove 	0	$\frac{V}{2}$

General fact (‘Folk theorem of evolutionary game theory’) **:** The replicator equation with fitnesses given by the payoffs of an evolutionary game *always* predicts evolution towards the ESS of the game.

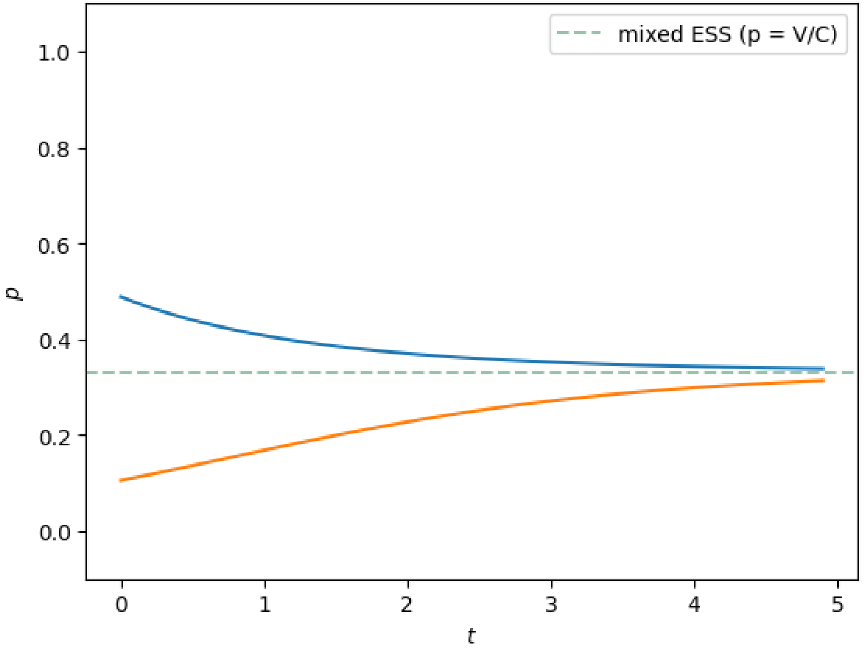
Let $p :=$ proportion of Hawks in the population
$1-p =$ proportion of Doves in the population

$$\text{Avg fitness of Hawks} \;= p\left(\frac{V-C}{2}\right)+(1-p)V$$

$$\text{Avg fitness of Doves} \;= (1-p)\frac{V}{2}$$

Recall from yesterday:

The **Replicator equation** $p(t+1) = \frac{W_H}{\overline{W}}p(t)$



Summary

Hawk-Dove*: A simple game to analyse whether ritualized displays for resource competition can evolve via natural selection. This model is very simplistic, uses many assumptions on ecological dynamics, and makes no quantitative predictions. However, the qualitative predictions can be insightful for clarifying logic. This is a general theme in evolutionary game theory models.

Opponent → / Focal individual ↓	Hawk	Dove
Hawk	$\frac{V-C}{2}$	V
Dove	0	$\frac{V}{2}$

Qualitative predictions of the simple model:

- If the resource is valuable relative to costs of injury ($V > C$), unrestrained aggression using weaponry (Hawk) is an ESS, and displaying without getting into a fight (Dove) is unstable.

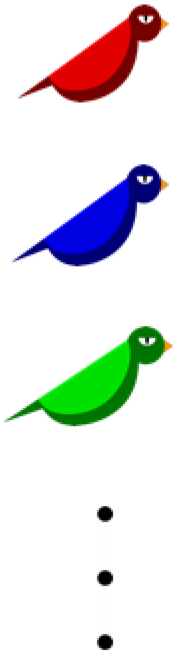
- If the costs of injury are high relative to value of the resource ($V < C$), ritualized display can be present as a mixed ESS due to natural selection. A proportion V/C of individuals should be Hawks, and 1-V/C should be Doves.

- The evolutionary equilibrium is *not* the configuration that is best for the population/species. Thus, evolution is not working for the good of the species in this model.

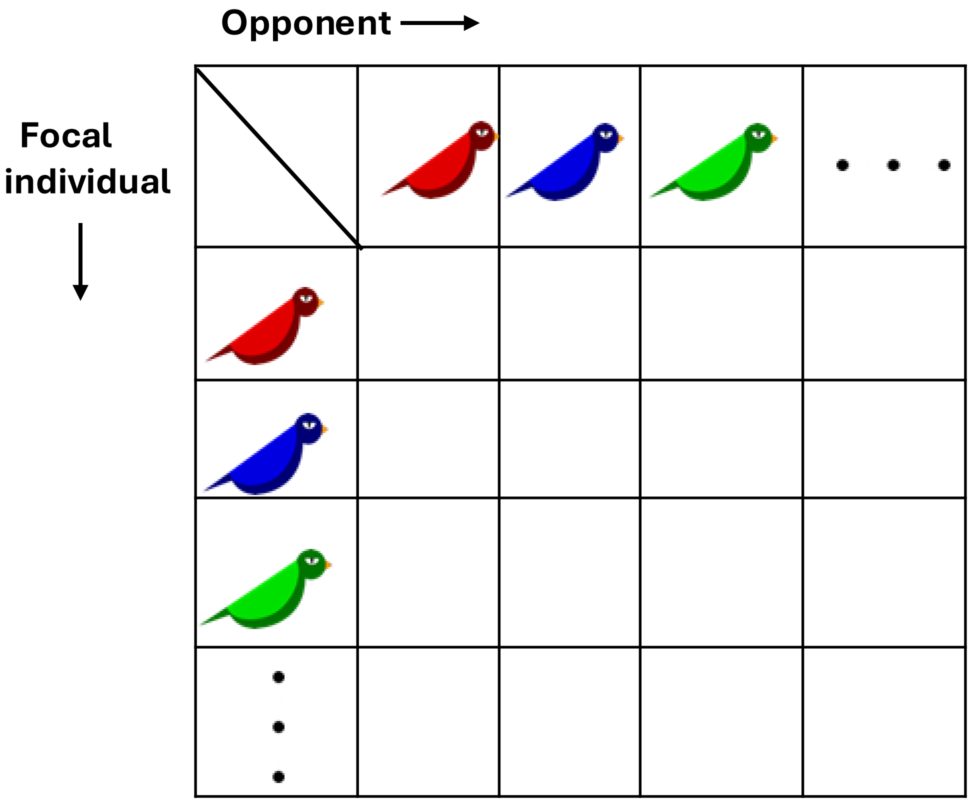
* This game is also called Chicken and Snow-Drift in the literature.

General recipe for making game theory models

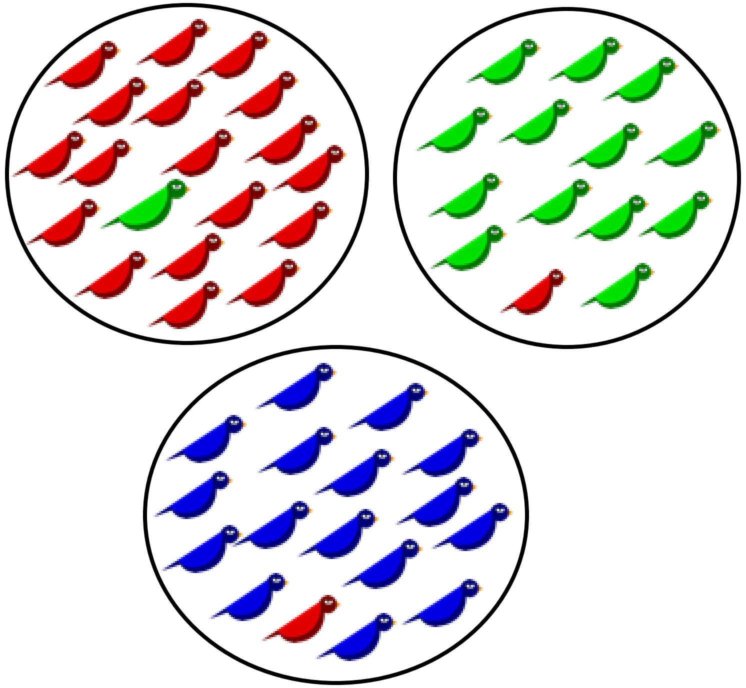
Step 1:
Define strategies of interest



Step 2:
Figure out how they interact



Step 3:
Find ESSs by invasion analysis



Some important extensions that we do not cover

‘Playing the field’

Anim. Behav., 1988, **36**, 416–432

Callers and satellites in the natterjack toad: evolutionarily stable decision rules

ANTHONY ARAK

Department of Zoology, Division of Ethology, University of Stockholm, S–106 91 Stockholm, Sweden

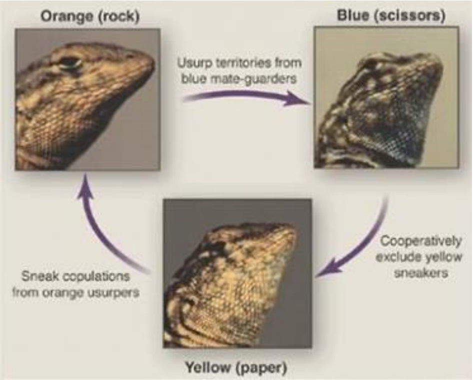
Cyclic dynamics

Letter | Published: 21 March 1996

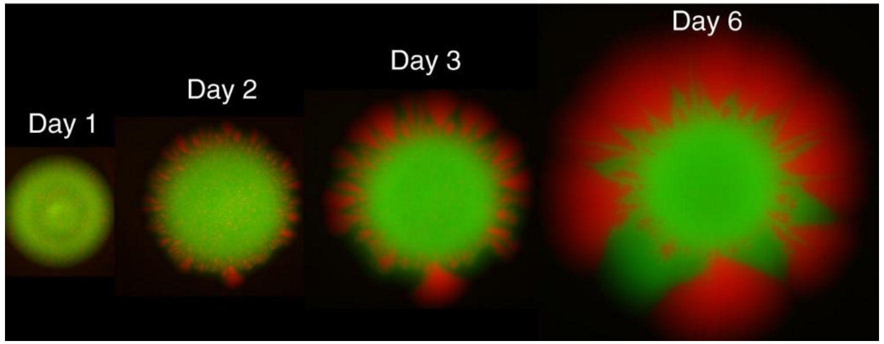
The rock–paper–scissors game and the evolution of alternative male strategies

B. Sinervo & C. M. Lively

Nature **380**, 240–243 (1996)



Spatial structure



REPORT · Volume 23, Issue 10, P919-923, May 20, 2013 · *Open Archive* Download Full Issue

Spatial Population Expansion Promotes the Evolution of Cooperation in an Experimental Prisoner’s Dilemma

J. David Van Dyken [1,2] · Melanie J.I. Müller [2,3] · Keenan M.L. Mack [4] · Michael M. Desai [1,2,3]

Affiliations & Notes Article Info

Letter | Published: 08 April 2004

Spatial structure often inhibits the evolution of cooperation in the snowdrift game

Christoph Hauert & Michael Doebeli

Nature **428**, 643–646 (2004) | Cite this article

Ownership asymmetries

Speckled Wood Butterfly *(Pararge aegeria)*

Anim. Behav., 1978, **26**, 138–147

TERRITORIAL DEFENCE IN THE SPECKLED WOOD BUTTERFLY (*PARARGE AEGERIA*): THE RESIDENT ALWAYS WINS

By N. B. DAVIES
Edward Grey Institute, Department of Zoology, Oxford

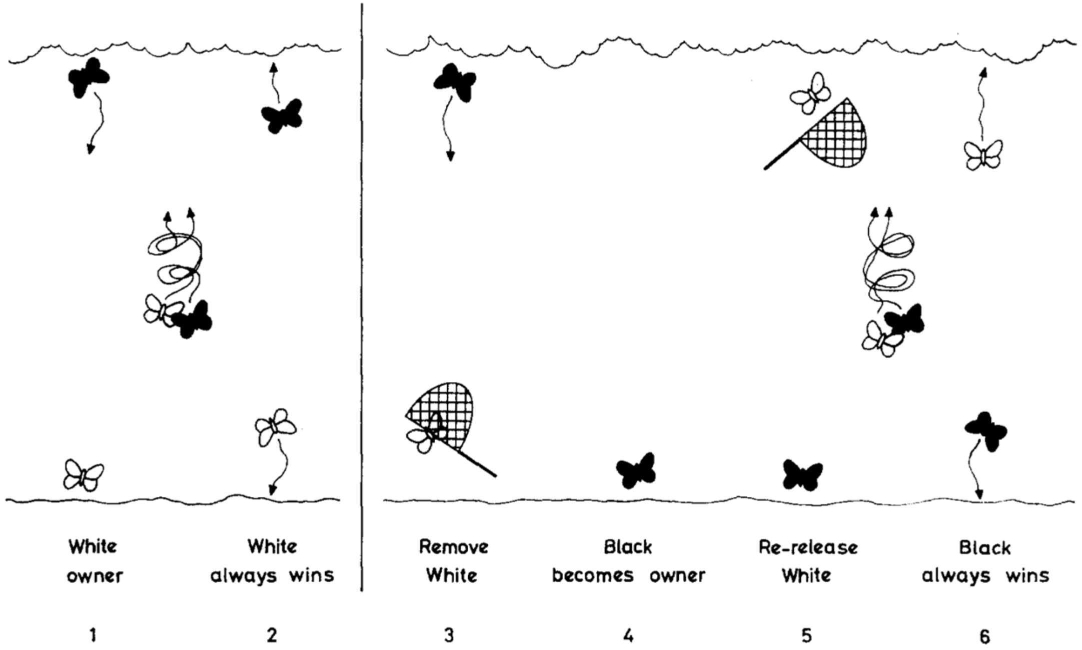
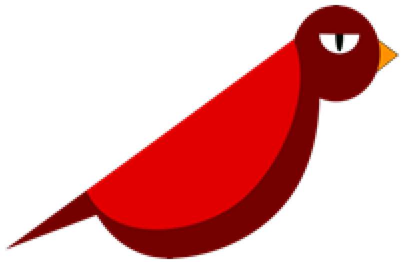


Table IV. Remove and Replace Experiments. When a Territory Owner is Removed from his Sunspot and then Released Again, He Loses His Territory if there is a Replacement Male Present, but Retains it if it is Empty

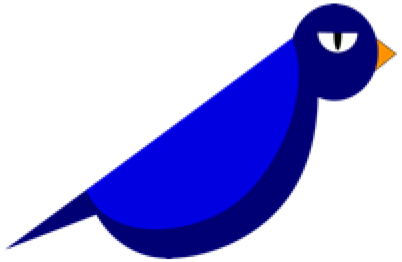
	No. of occasions the original owner: Retained territory	Lost territory
Replacement present	0	10
No replacement	10	0

Video from https://www.youtube.com/watch?v=Uwtm_B3YB38

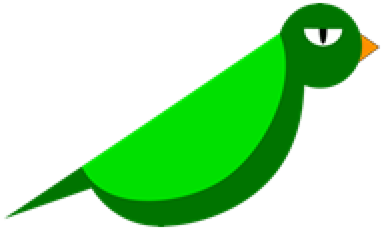
How to study this? Introduce a third strategy



“Hawk”: Aggressive, uses all its weaponry (beak, claws, fangs, venom, …), always fights and injures its opponent.



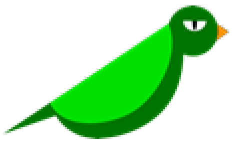
“Dove”: Engages in ritualised displays. If the opponent fights with weaponry, retreats and gives up the resource before any potential for injury.



“Bourgeois”*: If it arrived at the resource first, acts like a Hawk. If it arrived at the resource second, acts like a Dove.

* Because it believes in ownership of territory/land/property. I didn’t come up with this name, it was Maynard Smith 1979 *Proc. B.*.

Filling in the payoff matrix

Opponent → Focal individual ↓	Hawk 	Dove 	Bourgeois 
Hawk 	$\frac{V-C}{2}$	V	(hawk first: B acts like Dove) $\frac{1}{2} \cdot V$
Dove 	0	$\frac{V}{2}$	
Bourgeois 			

Filling in the payoff matrix

Opponent → / Focal individual ↓	Hawk 	Dove 	Bourgeois 
Hawk 	$\frac{V-C}{2}$	V	(B first: B acts like Hawk) $\frac{1}{2}\cdot V+\frac{1}{2}\frac{V-C}{2}$
Dove 	0	$\frac{V}{2}$	
Bourgeois 			

Filling in the payoff matrix

Opponent → / Focal individual ↓	Hawk 	Dove 	Bourgeois 
Hawk 	$\frac{V-C}{2}$	V	$\frac{1}{2} \cdot V + \frac{1}{2} \frac{V-C}{2}$
Dove 	0	$\frac{V}{2}$	(D first: B acts like Dove) $\frac{1}{2} \frac{V}{2}$
Bourgeois 			

Filling in the payoff matrix

Opponent → / Focal individual ↓	Hawk 	Dove 	Bourgeois 
Hawk 	$\frac{V-C}{2}$	V	$\frac{1}{2}\cdot V+\frac{1}{2}\frac{V-C}{2}$
Dove 	0	$\frac{V}{2}$	(B first: B acts like Hawk) $\frac{1}{2}\frac{V}{2}+\frac{1}{2}\cdot 0$
Bourgeois 			

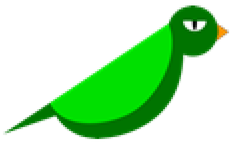
Filling in the payoff matrix

Opponent → / Focal individual ↓	Hawk 	Dove 	Bourgeois 
Hawk 	$\frac{V-C}{2}$	V	$\frac{1}{2}\cdot V+\frac{1}{2}\frac{V-C}{2}$
Dove 	0	$\frac{V}{2}$	$\frac{1}{2}\frac{V}{2}+\frac{1}{2}\cdot 0$
Bourgeois 	(B first: B acts like Hawk) (H first: B acts like Dove) $\frac{1}{2}\frac{V-C}{2}+\frac{1}{2}\cdot 0$		

Filling in the payoff matrix

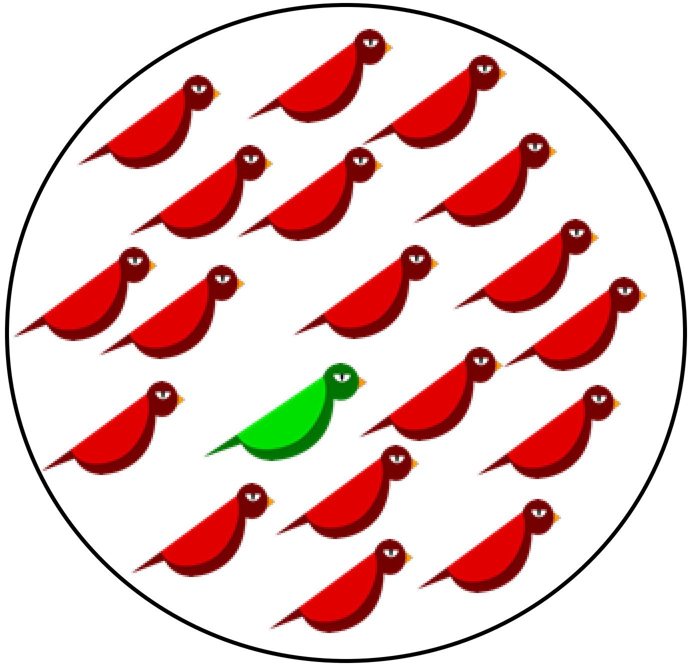
Focal individual ↓ \ Opponent →	Hawk 	Dove 	Bourgeois 
Hawk 	$\frac{V-C}{2}$	V	$\frac{1}{2}\cdot V+\frac{1}{2}\frac{V-C}{2}$
Dove 	0	$\frac{V}{2}$	$\frac{1}{2}\frac{V}{2}+\frac{1}{2}\cdot 0$
Bourgeois 	$\frac{1}{2}\frac{V-C}{2}+\frac{1}{2}\cdot 0$	(B first: B acts like Hawk) (D first: B acts like Dove) $\frac{1}{2}V+\frac{1}{2}\frac{V}{2}$	

Filling in the payoff matrix

Opponent → / Focal individual ↓	Hawk 	Dove 	Bourgeois 
Hawk 	$\frac{V-C}{2}$	V	$\frac{1}{2}\cdot V+\frac{1}{2}\frac{V-C}{2}$
Dove 	0	$\frac{V}{2}$	$\frac{1}{2}\frac{V}{2}+\frac{1}{2}\cdot 0$
Bourgeois 	$\frac{1}{2}\frac{V-C}{2}+\frac{1}{2}\cdot 0$	$\frac{1}{2}V+\frac{1}{2}\frac{V}{2}$	(Focal B acts like Hawk, opponent B acts like Dove) (Focal B acts like Dove, opponent B acts like Hawk) $\frac{1}{2}V+\frac{1}{2}\cdot 0$

Can Hawks be invaded by Bourgeois?

Focal individual ↓ \ Opponent →	Hawk	Dove	Bourgeois
Hawk	$\frac{V-C}{2}$	V	$\frac{1}{2}\cdot V+\frac{1}{2}\frac{V-C}{2}$
Dove	0	$\frac{V}{2}$	$\frac{1}{2}\frac{V}{2}$
Bourgeois	$\frac{1}{2}\frac{V-C}{2}$	$\frac{1}{2}V+\frac{1}{2}\frac{V}{2}$	$\frac{V}{2}$



$$\text{Hawk fitness} = \frac{V-C}{2}$$

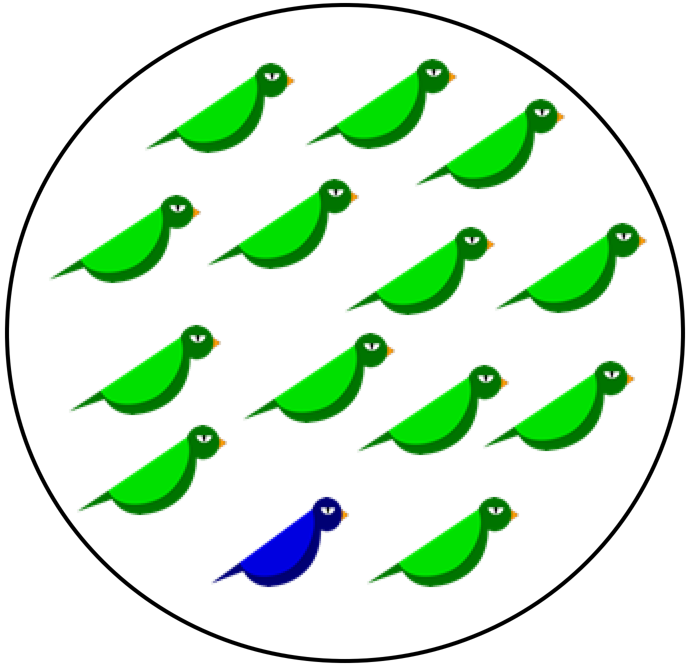
$$\text{Bourgeois fitness} = \frac{V-C}{4}$$

If $V > C$, Bourgeois cannot invade and Hawk remains an ESS.

If $V < C$, Bourgeois can invade a population of Hawks

Can Bourgeois be invaded by Doves?

Focal individual ↓ \ Opponent →	Hawk	Dove	Bourgeois
Hawk	$\frac{V-C}{2}$	V	$\frac{1}{2}\cdot V+\frac{1}{2}\frac{V-C}{2}$
Dove	0	$\frac{V}{2}$	$\frac{1}{2}\frac{V}{2}$
Bourgeois	$\frac{1}{2}\frac{V-C}{2}$	$\frac{1}{2}V+\frac{1}{2}\frac{V}{2}$	$\frac{V}{2}$



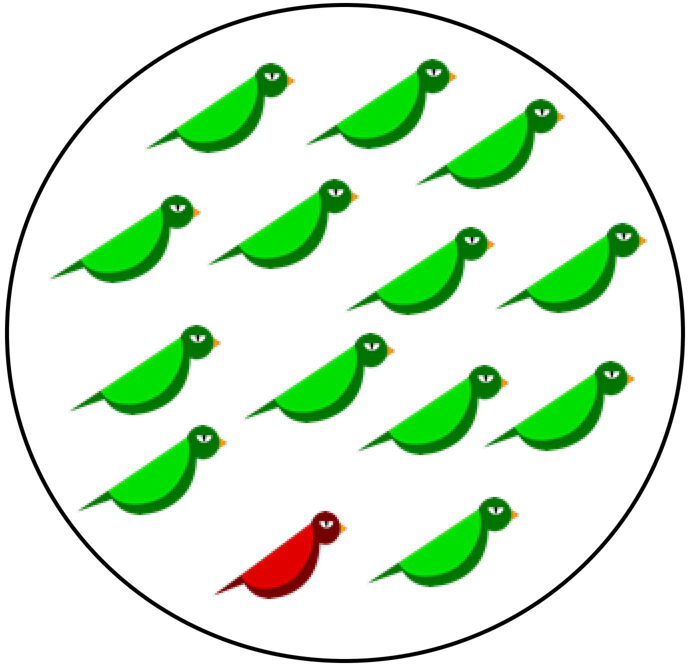
$$\text{Bourgeois fitness} = \frac{V}{2}$$

$$\text{Dove fitness} = \frac{V}{4}$$

No, Bourgeois is always stable against Dove

Can Bourgeois be invaded by Hawks?

Focal individual ↓ / Opponent →	Hawk	Dove	Bourgeois
Hawk	$\frac{V-C}{2}$	V	$\frac{1}{2}\cdot V + \frac{1}{2}\frac{V-C}{2}$
Dove	0	$\frac{V}{2}$	$\frac{1}{2}\frac{V}{2}$
Bourgeois	$\frac{1}{2}\frac{V-C}{2}$	$\frac{1}{2}V + \frac{1}{2}\frac{V}{2}$	$\frac{V}{2}$



$$\text{Bourgeois fitness} = \frac{V}{2}$$

$$\text{Hawk fitness} = \frac{V}{2} + \frac{V-C}{4}$$

Hawk can invade if $V > C$. If $V < C$, Bourgeois is an ESS.

When $V < C$:

Focal individual ↓ \ Opponent →	Hawk	Dove	Bourgeois
Hawk	$\frac{V-C}{2}$	V	$\frac{1}{2} \cdot V + \frac{1}{2} \frac{V-C}{2}$
Dove	0	$\frac{V}{2}$	$\frac{1}{2} \frac{V}{2}$
Bourgeois	$\frac{1}{2} \frac{V-C}{2}$	$\frac{1}{2} V + \frac{1}{2} \frac{V}{2}$	$\frac{V}{2}$

Optional exercise if you are curious:

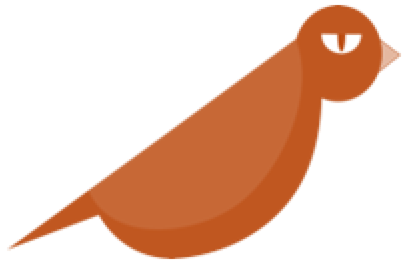
check whether Bourgeois can invade the Hawk-Dove mixed ESS we derived earlier (*V/C* Hawks, *1-(V/C)* Doves)

- Bourgeois is an ESS (cannot be invaded by either hawk or dove).
- Bourgeois can invade a population of Hawks.
- Bourgeois can invade a population of Doves.

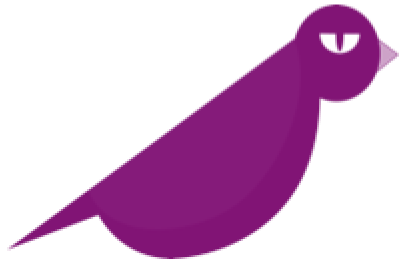
Notice: The act of playing Hawk if you arrive *first* is arbitrary. A strategy where you play Dove if you arrive at the resource first and play Hawk if you arrive second* would have the exact same payoffs. Thus, the actual rule is arbitrary as long as B-B interactions agree on the rule. Such arbitrary asymmetries are called ‘conventions’.

* Maynard Smith 1979 *Proc. B.*. calls this strategy ‘anarchist’.

Some other strategies to try in the Hawk-Dove game



“Retaliator/Reciprocator”: Acts like a Hawk towards Hawks. Acts like a Dove towards Doves. (You decide what happens when two Reciprocators meet)



“Bully”: Acts like a Hawk towards Doves, showing aggression. If the opponent is a Hawk, instead behaves like a Dove and runs away.

P.S: Which combination is best for mean fitness?

Opponent → Focal individual ↓	Hawk 	Dove 
Hawk 	$\frac{V-C}{2}$	V
Dove 	0	$\frac{V}{2}$

Let $p :=$ proportion of Hawks in the population
$1 - p =$ proportion of Doves in the population

$$\text{Avg fitness of Hawks } = p\left(\frac{V-C}{2}\right) + (1-p)V$$

$$\text{Avg fitness of Doves } = (1-p)\frac{V}{2}$$

$$\text{Mean fitness } \overline{W} = \text{Prop. Hawks} \times \text{Fitness of Hawks} + \text{Prop. Doves} \times \text{Fitness of Doves}$$

$$= p\left[p\left(\frac{V-C}{2}\right) + (1-p)V\right] + (1-p)^2\frac{V}{2}$$

$$= \frac{V}{2} - \frac{C}{2}p^2$$ (you can trust me on this if you want)

It is best for the *population* if $p = 0$ (all Doves). However, as we saw earlier, all Doves is not evolutionarily stable.